

CSE 318 — Online 2

Section C2

In your semi-greedy heuristic, after building the RCL, you pick a vertex **uniformly at random** from it. This means the best and worst candidates in the RCL are equally likely to be chosen.

Task: Change the selection so that **better candidates are more likely to be picked**. Specifically:

1. Build the RCL exactly as before (value-based, using α).
2. For each v in the RCL, compute $g(v) = \max\{\sigma_X(v), \sigma_Y(v)\}$.
3. Instead of picking uniformly, pick vertex v with probability proportional to $g(v)$:

$$P(v) = \frac{g(v)}{\sum_{u \in \text{RCL}} g(u)}$$

4. If all g values in the RCL are zero (or all equal), fall back to uniform random selection.
5. Place the selected vertex using the same greedy placement rule as before.

Everything else stays the same — RCL construction, seeding, local search, GRASP loop.

What to run and report:

- Run GRASP with biased selection on at least 6 benchmark graphs, using $\alpha = 0.3$ and $\alpha = 0.7$.
- Run your original GRASP (uniform selection) on the same graphs with the same α values.
- Report the best cut value found for each.
- Make a comparison table.